DYNALYTICS: THE BIRTH OF A FRAMEWORK FOR RIGOROUS ANALYTICS OF DYNAMICAL SYSTEMS

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ABSTRACT

In this paper, we introduce *Dynalytics*, an overarching framework and methodological ecosystem designed for the analysis and rigorous validation of dynamics of and dynamical systems. The framework encompasses a diverse family of models, including transition networks, co-occurrence networks, psychological networks, and higher-order networks. Central to *Dynalytics* is the application of a multi-level confirmatory testing battery to validate the supported models and support any analytical claim: split-half reliability for internal consistency, bootstrapping for edge-level stability, case dropping for centrality stability, and permutation-based group comparisons. The framework is supported by an open-source software ecosystem including R, Python, and Javascript libraries, as well as web and desktop applications. More importantly, *Dynalytics* is grounded in a "scientific contract" in which every analytical claim must be matched by evidence appropriate to the structure, scope, and complexity of the inference being made.

Keywords Dynalytics · Network Analysis · Process Analysis · Complex Dynamic Systems · Transition Network Analysis

1 Introduction: From Transition Network Analysis to *Dynalytics*

Research on learning processes has a long and methodologically rich history of contributions from multiple disciplines and evolving analytical traditions. Scholars have drawn on a wide spectrum of methods to understand how the learning process progresses over time, ranging from in-depth qualitative methods—such as ethnography, discourse analysis, and case studies—to computational and data-driven methods like network analysis, process mining, and sequence analysis [1]. Each of these approaches offers a distinct lens that allows the modeling of patterns, structures, and temporal dynamics across diverse contexts and learning scenarios. Over time, this interdisciplinary exchange has generated a substantial body of knowledge, with methods often adapted and appropriated from other fields to address the complexity of learning processes [1]. However, while the appropriation of methods from other disciplines has been productive, it has often been accompanied by limitations in scope, contextual alignment, and theoretical grounding. In many cases, methods developed for different phenomena are hard to align with the nature of learning, resulting in analyses that are methodologically sophisticated but insufficiently attuned to the theoretical foundations of learning processes and the learning sciences.

Take, for instance, process mining, which aims at discovering, describing, and optimizing processes based on event logs [2]. While useful for identifying workflows and deviations from expected paths, it is primarily oriented toward conformance to predefined models and efficiency of execution of business processes. As such, it often lacks a strong inferential framework and may impose normative process structures on learning interactions that are inherently agentic, emergent, and context-sensitive [3]. Other methods, such as Social Network Analysis (SNA), have been extensively used to model relationships and interactions among learners, teachers, and peers [4, 5]. Yet, SNA's theories and methods were designed to capture who connects with whom and how those connections shape behaviour—a fundamentally different question from how a learning process unfolds [4]. While researchers have successfully adapted SNA, its statistical validation methods were hard to apply to the learning processes—where the actors are not learners, e.g., steps or actions of a process—that are essentially temporal, interdependent, and follow a non-social structure.

In the grand scheme of things, our research landscape, theoretical or empirical claims about the processes are frequently based on observations, with limited assurance that these patterns would withstand rigorous statistical validation or replicate across contexts. Such a foundation is not conducive to either robust research or reliable practice. For findings to inform practice, policymakers and educators need confidence that identified patterns are stable, generalizable, and not artifacts of specific samples, coding decisions, or analytical choices.

Transition Network Analysis (TNA) [6] was introduced to fill this gap by providing a principled, multi-level confirmatory framework—spanning several methodologies that validate the process as a whole and its structure, the individual transitions that make up the fabrics of the learning process, the patterns, the comparisons between them, as well as the important events that shape the process. The framework embeds extensive confirmatory testing procedures that ensure analytical findings reflect genuine system behaviour rather than noise or chance. Entire network models can be assessed through null-model testing [7], split-half reliability [8], and bootstrap analysis [9]. Individual edges can be examined using bootstrap procedures, and centrality measures can be evaluated through case-dropping techniques [9]. Comparative analyses benefit from resampling strategies, permutation approaches, and graph-theoretic significance assessments, all designed to substantiate the validity of observed patterns. In this way, TNA functions not only as a conceptual guide to dynamic systems analysis but also as a rigorously validated methodological ecosystem dedicated to producing reliable, interpretable, and scientifically robust insights.

Whereas some of these statistical tests have been used before either in other fields (e.g., psychology) or traditional psychometrics, the contribution of TNA lies in adapting and validating these tests using vast datasets and robust simulation studies. More importantly, TNA has set the foundation and the methodological formalism of these tests, defining their scope, underlying assumptions, statistical procedures, and appropriate application boundaries.

To achieve these goals, TNA development followed several stages. First, we conducted a comprehensive review of validation approaches in education and in adjacent fields that have recently advanced network modeling, particularly psychological networks and network science. Second, each verification method was rigorously examined against its underlying assumptions, with particular attention to their suitability for sequential and process-oriented data. This was followed by systematic validation of each test using established methods such as the use of simulated data with known properties to assess whether TNA procedures could accurately recover expected patterns. In addition, both empirical checks and formal mathematical scrutiny were applied to audit the behavior, boundaries, and reliability of the proposed methods. As such, each statistical method or test in TNA has been carefully calibrated, rigorously validated, and subjected to an extensive battery of simulation-based evaluations, empirical checks, and formal analytical scrutiny to ensure its reliability, robustness, and appropriate fit of application.

Nevertheless, the conceptual groundwork of TNA testing battery and confirmatory framework applies broadly to networks derived from event data or estimated from empirical observations—in other words, to networks constructed from observed interactions, sequences, or co-occurrence processes where structure emerges from underlying data-generating mechanisms rather than being assumed a priori or observed [6]. Whether applied to social, behavioral, or social systems, the testing framework offers a unified lens through which inferences can be examined—capturing how patterns form, stabilize, transform, or characterize different groups, to mention a few. This expansive scope underscores TNA's dual contribution. Its value transcends the narrow focus on the analysis of transitions, extending instead to providing a unified theoretical and methodological framework for describing change and stability in any dynamic system.

Dynalytics is the name that we use to refer to this overarching framework, which encompasses TNA and the diverse family of models and statistical methods that use the same unified philosophy of scientific rigor and a consistent confirmatory testing battery.

2 Dynalytics Promise and Philosophy

Dynalytics is built and grounded in the "scientific contract" principle in which every analytical claim must be matched by evidence appropriate to the structure, scope, and complexity of the inference being made. The framework (outlined in Fig. 1) assumes that methodological validity is established through the alignment between the substantive claim, the underlying data-generating process, the representational model, and the confirmatory procedures used to evaluate it. As such, inference is treated as conditional on evidence based on strong and appropriate statistical foundations.

In *Dynalytics*, different forms of interpretation require appropriate forms of validation. Claims concerning the reliability or structure of an entire network require evidence of structural reproducibility and stability [10]. Claims about specific transitions require uncertainty estimation at the edge level. Claims regarding influential states require demonstrations that centrality structures remain stable under perturbation and resampling. Comparative claims between groups, conditions, or temporal phases require permutation-based inference and structural comparison procedures capable of distinguishing genuine reorganisation from stochastic variation.

The same logic extends to temporal, heterogeneous, and higher-order phenomena. More importantly, these standards extend to other forms of analysis supported by the framework that are not essentially network models. For instance, claims about sequential dependency require evidence that the assumed memory structure is supported by the data-generating process. Claims regarding latent subgroups or process typologies require clustering or mixture-based evidence. Claims involving polyadic, multilayer, or higher-order interaction structures require representational frameworks capable of preserving those dependencies rather than reducing them to pairwise approximations. Mechanistic claims about behavioural or cognitive processes require pathway and pattern evidence that reconnects network structure to the trajectories from which it emerged.

The framework adopts a proportional and progressive view of scientific evidence: the complexity of the evidential standard must match the complexity of the claim. Simple descriptive observations require descriptive support, whereas mechanistic, comparative, or structural interpretations require stronger forms of confirmatory evidence. *Dynalytics* therefore treats estimation, validation, robustness assessment, and interpretation as inseparable components of a single inferential system designed to ensure that conclusions remain faithful to the empirical process that generated the data. All the more so, conclusions must be supported by evidence proportionate to the scope, explanatory depth, and inferential demands of the claims being advanced.

3 Estimation Methods

Analysis, estimation, and the production of evidence represent the cornerstone of the method's operationalization and the primary mechanism through which empirical observations are transformed into structured, interpretable, and scientifically defensible conclusions. For instance, network estimation defines the empirical object under study and encompasses a wide array of approaches suited to different analytical questions. The research question, the assumed data-generating mechanism, and the theoretical orientation jointly determine which estimation strategy is appropriate, and this choice carries substantive implications for both the resulting structure and the conclusions drawn from it. A transition network captures how events unfold through directed temporal dependencies, whereas a co-occurrence network reflects associative proximity without implying sequence or temporal ordering [6]. An attention network, in turn, models multi-step influence and the propagation of activity or focus across interconnected elements [11]. Although these networks may appear structurally similar at the level of graph representation, they rest on different assumptions regarding what constitutes a relation, how dependency is operationalized, and which processes the network is intended to represent. A wide array of analytical methods exist across several types of data and empirical questions that go beyond networks contributing to a rich toolset of analytical methods.

Estimation or analysis are the entry point to inference, and the choice of method constrains every downstream interpretation. The framework is organised around a common registry of named models, so a method is added or replaced without altering the surrounding inferential machinery. A study that progresses from TNA to a Higher-Order Network, or from a co-occurrence network to a simplicial complex, inherits the confirmatory approach e.g., bootstrap, permutation, reliability, and stability tools, and the named-model discipline keeps the analyst, the reader, and the software in agreement about which object is under examination at each stage.

Within the *Dynalytics* framework, each approach is therefore treated as a distinct model in which the estimator and the resulting output form a unified methodological entity. Different estimators operationalize different forms of dependency, temporal structure, stochastic assumptions, and relational meaning, allowing the same empirical data to generate analytically distinct representations of the underlying system. At the same time, these approaches benefit from the broader confirmatory and validation architecture established through Transition Network Analysis [12]. Regardless of the specific model employed, results can be examined through a rigorous ecosystem of estimation, robustness assessment, inferential testing, and comparative evaluation designed to ensure that observed structures reflect genuine system behaviour rather than stochastic variation or methodological artefacts. Consequently, *Dynalytics* provides a common methodological foundation through which heterogeneous models can be examined under consistent standards of reliability, validity, and interpretability while preserving the theoretical and analytical specificity of each modeling approach. As we currently stand, the main methods supported by the framework are described in the next section.

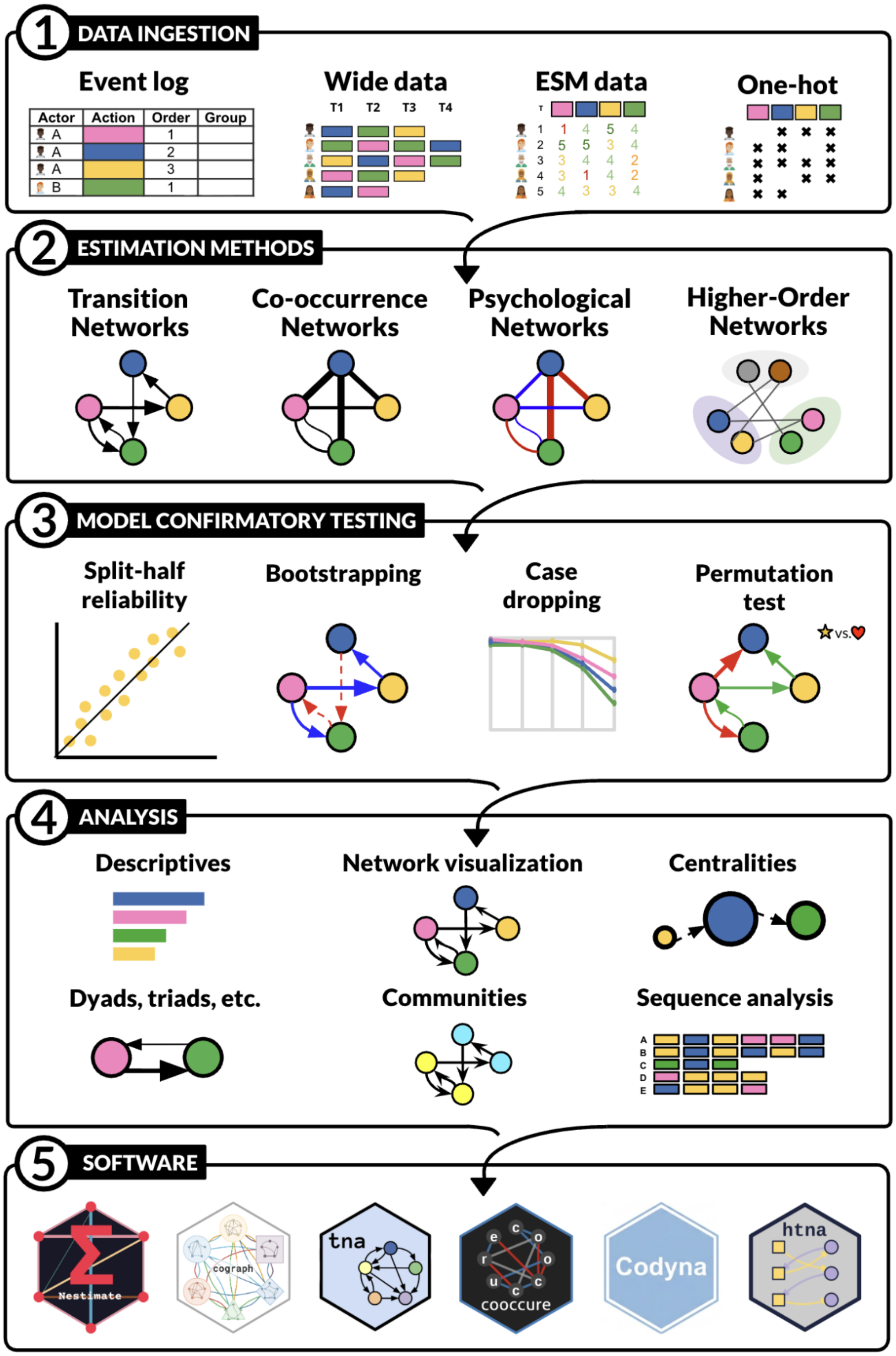

Figure 1: Overview of Dynalytics' layers.

3.1 Transition Network Family

TNA models estimate directed transition probabilities from sequence data. Each edge expresses the conditional probability of a target state given a source state, and the network is a Markov model of order one over the observed states [6]. TNA answers questions about sequential dynamics: which actions or states tend to follow which prior actions, which behavioural states lead to later states, and how a process unfolds step by step. Relatedly, Frequency Transition Network Analysis (FTNA) [13] preserves the raw count of observed transitions rather than their row-normalised proportions based on Markov models like TNA. The resulting network captures empirical frequencies. The distinction from TNA matters for interpretation. A transition can have a high conditional probability under TNA because its source state is rare, and a different transition can have a lower probability while accounting for a much larger share of the observed process under FTNA. TNA describes relative tendency; FTNA describes observed frequencies. Both are estimated from the same sequence data, and the choice between them is a choice of a substantive question [13]. Attention-based Transition Network Analysis (ATNA) [11] captures the influence across the timeline using a multi-step approach that addresses the limitations of limited memory models, such as TNA. Earlier events continue to shape later ones.

Heterogeneous Transition Network Analysis (HTNA) [14] models sequences in which two or more actor types share a single process—for example, human and AI agents within a session, multiple participant roles within a dialogue, or different categories of nodes within an interaction system. Heterogeneity also extends to the nature of the edges themselves. In this sense, HTNA can be viewed as a generalization of bipartite network structures, in which multiple node classes coexist within the same graph, but differs by allowing edges to form both within and across node types, thereby supporting transition structures that capture sequential dynamics among heterogeneous actors. Meta-path analysis identifies actor-level pathways such as human $\rightarrow$ AI $\rightarrow$ human sequences. Relatedly, Multicluster Multilayer Networks (MCML) preserve layered organisation when the data are stratified by group, cluster, time slice, or actor. MCML stores within-layer transition structure together with the between-layer aggregate, which keeps comparisons across strata available without collapsing the data into a single matrix.

3.2 Co-occurrence networks

Co-occurrence Network Analysis (CNA) represents associations (instances where the state or action co-occur, happen, or are associated with each other) within an observational unit—a session, a window, a person, a document [15]. The relation is associative rather than sequential. Window-based Transition Network Analysis (WTNA) makes use of binary data (data coded as 0s and 1s) and computes transitions and co-occurrences inside local temporal neighbourhoods of fixed size. Furthermore, WTNA makes use of both the fact that events co-occur together in the same window and also that each window follows the other. WTNA are thus mixed network models with both types of interactions in the same network, which is a novel and innovative approach to taking the full advantage of the data.

3.3 Beyond traditional networks

For relations that cannot be reduced to dyads, the *Dynalytics* framework offers further analysis types. In other words, *Dynalytics* is not constrained, restricted or dogmatically attached to networks, or a certain type of models. Therefore, *Dynalytics* include novel models that go beyond its foundational approaches. Hypergraph networks represent polyadic relations directly through hyperedges that connect more than two nodes, with hypergraph centralities and hypergraph structural measures that respect the polyadic geometry. Simplicial complex networks represent nested relations among sets of nodes. Namely, *Dynalytics* includes rich functions for Higher-Order Networks (HON, HONEM, HYPA, MOGen) [16], which encode explicit temporal memory and account for multiple dependencies, structural complexities, and cases where interaction patterns cannot be reduced to simple pairwise steps (e.g., simplicial complexes [17]). Put another way, they are designed for situations in learning where what a learner does next depends on the sequence of prior actions rather than only the most recent one. We expect that HON will have extensive applications in the analysis of complex processes, structures, and human-AI interactions and therefore, they make an important part of the Nestimate R package [18].

3.4 Psychological networks

The *Dynalytics* framework extends to association methods for association, ecological, and panel data, again named in correspondence with the network each method produces. Correlation networks describe pairwise linear association. Partial correlation networks condition each pairwise association on the remaining variables and recover the direct-dependence structure after controlling for shared variance [19]. Gaussian graphical models estimated by graphical lasso (EBICglasso) add L1 regularisation under the extended Bayesian information criterion, producing sparse partial-correlation networks suited to high-dimensional data [20]. Ising networks fit the same logic to binary data through L1-penalised nodewise logistic regression. Mixed graphical model networks (MGM) extend nodewise regression to data that combine continuous and categorical variables, with EBIC lambda selection and Lin–Wei thresholding. Multilevel vector-autoregression networks (mlVAR) estimated from intensive longitudinal data return three named networks at distinct time scales—a temporal network of fixed-effect lagged coefficients, a contemporaneous network of within-person residual partial correlations, and a between-person network of partial correlations among person means. GIMME networks apply group-iterative multiple-model estimation to subject-specific connectivity in fMRI and ESM data [21]. All of such models are offered through the network estimation main network approach Nestimate [18].

4 Confirmatory Testing

The *Dynalytics* framework relies on a rigorous and strict approach to validation. The estimated models must be examined for internal reproducibility, sampling uncertainty, robustness to information loss, comparability across groups, and adequacy of their temporal assumptions. The confirmatory layer of the framework operationalises each of these requirements through a specific statistical procedure rooted in an established scientific principle according to the methods used.

4.1 Model suitability

Dynalytics incorporates a suite of diagnostic tools to determine model suitability, beginning with an evaluation of the memoryless assumption. Markov-order adequacy testing provides the empirical basis for this assessment and examines whether a first-order transition structure sufficiently represents the system or if the underlying process necessitates modeling longer temporal dependencies [22]. On the one hand, a first-order model is deemed appropriate; TNA would be appropriate. On the other hand, if higher-order dependencies emerge, another type of model is then necessary that attends to and preserves the structural fidelity to the data generation process e.g., ATNA, FTNA or HON. Following the determination of model order, path-dependence diagnostics pinpoint the specific loci where temporal memory influences the system.

4.2 Model reliability

To validate model reliability, we rely on split-half reliability, which draws its logic from classical test theory [23]. A reliable instrument should yield similar scores when the items are partitioned into two equivalent halves. Translated to network estimation, the instrument is the estimator, and the score is the matrix of edge weights. Cases are randomly partitioned, two networks are estimated independently, and the matrices are compared through edge-wise correlation and absolute deviation. Repeated splits provide a distribution of agreement values. The test answers whether the estimated network is internally reproducible within the available data. Low reliability indicates that the network is too sample-dependent to support strong interpretation. Split-half reliability does not address external generalisation; it addresses internal consistency.

4.3 Edge-level stability

Bootstrap analysis evaluates edge uncertainty through resampling procedures designed to ensure that estimated parameters genuinely reflect the underlying data-generating process [9]. Resampling preserves the inherent temporal dependencies and structural fidelity of the system to ensure that uncertainty estimates capture authentic variability rather than methodological artefacts. This confirmatory approach allows researchers to verify that individual edges are statistically consistent rather than spurious reflections of noise. Consequently, bootstrap procedures provide the empirical substantiation necessary for validating theoretical assumptions and constructing robust models that could remain reproducible across diverse datasets.

Other methods of examining edge stability include edge-weight case-dropping stability, which evaluates robustness to information loss [9]. The procedure repeatedly removes increasing proportions of cases, re-estimates the network, and compares the reduced-sample edge vector with the full-sample vector through rank correlation. High edge stability supports interpretation of the network as a stable map of relations; low edge stability indicates that the structure is fragile and should be reported as sample-specific. Usually, both methods agree on the strong, stable edges, which provide evidence of these stable, consistent structures across different stability methods.

In the same way, centrality stability utilize case-dropping procedures to verify the structural significance of key states or events. Generally, researchers use centrality to advance theoretical assertions—characterizing a state as either a bridge, an influential hub, or a peripheral event. These inferences demand empirical proof that structural rankings persist even when information loss is introduced. Therefore, in *Dynalytics*, centrality analysis entails a dual process: estimation of the centrality measure and stability analysis confirms whether that position reflects a genuine system property capable of supporting substantive interpretation.

4.4 Model comparison

Network analysis is often used to compare groups, sessions, or subpopulations such as high and low achievers, yet many of these comparisons remain descriptive, visual, or subjective in nature [24]. *Dynalytics* addresses this limitation by providing a structured inferential framework for network comparison that is grounded in formal statistical testing rather than visual inspection alone. Network comparison testing extends permutation to structural hypotheses through a set of complementary statistics. *Dynalytics* provides a multi-level network comparison framework for contrasting transition networks across groups, conditions, or time points. At the edge level, the focus is on identifying which specific transitions differ between networks. At the summary level, overall similarity is assessed using correlation, dissimilarity, and deviation-based measures. At the centrality level, comparisons examine whether structural roles of nodes are preserved or redistributed across networks. At the network level, global properties such as density, centralization, and reciprocity are contrasted to assess broader structural organization. Each level is accompanied by permutation-based inference to ensure that observed differences can be distinguished from variation expected under chance, and thereby grounding comparative claims in a coherent statistical framework.

Permutation testing provides inference under exchangeability [25]. Under the null hypothesis of no group difference, group labels are arbitrary. If cases are exchangeable across groups, shuffling the labels generates a valid null distribution for any function of the two networks—global strength, edge-wise differences, centrality differences, pattern frequencies. The observed difference is then located within this empirical null and assigned a p-value.

Permutation testing is also distribution-free in the same sense as the bootstrap: it dispenses with a parametric distribution for the test statistic, but it requires that the permuted units are genuinely exchangeable under the null. Designs with paired or clustered structure must preserve that structure during shuffling. The permutation test is the natural device for group comparison in the framework, because two networks estimated separately almost always look different when plotted side by side, and the relevant question is whether the observed difference exceeds what arises by chance under exchangeability.

4.5 Other confirmatory methods

The *Dynalytics* framework includes several other analytical methods and tools that, for instance, estimate the transition entropy and quantify the dispersion of outgoing transitions from a state and help establish the type of model needed. Markov stability evaluates whether the modular structure persists across scales of a transition process. Passage-time analysis estimates the expected number of steps required to reach one state from another through the transition process. It extends interpretation beyond direct edges to eventual reachability. A state can be dynamically close to an outcome through a chain of intermediate steps even when the direct edge between them is weak. Passage-time analysis supports claims about accessibility, gateway states, and temporal distance that the immediate adjacency matrix cannot articulate.

5 The Analysis Layer

The models in *Dynalytics* benefit from a unified analysis layer that operates consistently across estimation types. The analysis layer sits between model estimation and interpretation, and it applies through a shared system in which all estimators return a common structure. Several families of operations define the *Dynalytics* analysis layer: descriptives, centrality, community detection, visualisation, and pattern analytics, among others. These functions apply widely and can be implemented across TNA, FTNA, CNA, ATNA, WTNA, HTNA, and higher-order networks when appropriate.

Descriptive analysis characterises the empirical foundation of the model. State frequencies, base rates, initial and terminal distributions, and transition structure provide the contextual grounding for interpreting edges, centrality, and group differences. Descriptive measures define the distributional landscape on which all subsequent inference depends. In other words, these descriptive measures offer the base backgrounds upon which interpretation relies and widen the picture to different perspectives of the data.

Centrality analysis captures structural roles within the network. Measures such as in-strength, out-strength, betweenness, closeness, and eigenvector-based influence define and measure the importance of events or states according to the network model and operationalisation of the data [5]. Community detection identifies modular structures within the network [26]. Communities can be interpreted as functional or thematic groupings depending on the domain, and are used for cross-scale and cross-group comparisons.

Visualisation and network comparison, together with bootstrap plots, help communicate structure and provide a summary view of the data in an intuitive form. Because plotting is decoupled from estimation, similar plots can be rendered across multiple visual modalities without altering their inferential content.

Sequence-pattern analytics recover and mine patterns that are not easily visible through inspection. *Dynalytics* offers a wide array of pattern mining methods [27], e.g., n-gram and gapped-pattern mining, motif detection, association rules, sequence patterns, and pattern–outcome models, reconnecting network structure to the underlying trajectories that generated it. This layer moves analysis from dyadic associations to temporally extended mechanisms, linking local transitions to higher-order behavioural or cognitive sequences.

Dynalytics integrates clustering methods that group individuals based on similarity in their data [28], identifying distinct behavioural profiles or process typologies within a population. Pattern mining operates within and across clusters to characterise the sequential signatures that distinguish each group, revealing not only that individuals differ but also how their temporal dynamics differ in terms of specific transitions and sequential patterns. *Dynalytics* supports the integration of external covariates with cluster assignments, enabling researchers to examine how individual characteristics—such as achievement, demographics, prior knowledge, or motivational profiles—relate to membership in behaviourally defined clusters. This bridges the gap between person-level variables and process-level typologies, allowing researchers to test whether theoretically meaningful covariates predict distinct patterns of temporal behaviour.

5.1 Validation of *Dynalytics* Analysis Routines

Dynalytics follows a multistructure strict equivalence and validation policy to ensure that every statistical procedure is both theoretically correct and empirically faithful to the data-generating process. Each *Dynalytics* method is required to implement the exact statistical question it is intended to answer, including correct resampling units in bootstrap procedures, correct exchangeability assumptions in permutation tests, correct conditional-independence structures in Markov-order tests, and correct re-estimation logic in stability and higher-order analyses. This alignment is enforced through a strict numerical equivalence framework.

Validation is carried out through four complementary strategies that together follow current software-engineering best practice: numerical agreement, generative calibration, input/data validation, and behavioural testing of happy and unhappy paths. First, each implementation is benchmarked against established R and Python packages to ensure that outputs match reference results within extremely tight numerical tolerances. Second, and more fundamentally, Dynalytics tests are assessed on simulated data where the true data-generating process is known, allowing direct evaluation of whether each test recovers the correct structures, parameters, and model choices under controlled conditions; transition-network estimators, for instance, have been assessed on millions of datasets spanning diverse configurations, sample sizes, sequence lengths, and missing-data regimes, and bootstrap stability procedures have demonstrated near-perfect sensitivity and very high specificity in identifying true and false transitions with predictable performance across a wide array of variations. Third, every public function performs explicit data validation at entry—checking types, shapes, admissible ranges, identifier consistency, and temporal ordering—so that malformed inputs fail fast with informative errors rather than producing silently incorrect results. Fourth, the test suite covers both happy paths (well-formed inputs under canonical assumptions) and unhappy paths (missing data, ties, empty sequences, singleton trajectories, degenerate transition matrices, non-stationary inputs, and adversarial edge cases), ensuring that estimators degrade gracefully and that error states are themselves part of the verified contract. Continuous integration runs the full suite on every change, so each release inherits the same guarantees rather than relying on ad-hoc verification.

Across all other tests, including permutation-based comparison tests, centrality stability measures, Markov-order selection, mixed graphical models, and higher-order estimators, followed the same dual validation mechanisms to reproduce known reference implementations and correctly recover known ground-truth structures in simulated data. Only after passing both criteria—numerical equivalence and generative recovery—are procedures considered valid for use. This makes validation an integral part of the methodological framework rather than a post hoc check, ensuring that every implemented test is both computationally correct and substantively meaningful.

6 Software Implementations

Dynalytics is implemented as a layered ecosystem of open-source software packages designed to separate estimation, analysis, and visualisation while maintaining a unified analytical workflow. The R ecosystem serves as the reference implementation, complemented by an ecosystem for browser-based and desktop applications. Across both ecosystems, packages share common data structures and interoperable object schemas, allowing methods to operate consistently across estimation types and analytical contexts. More importantly, a rigorous validation process ensures that these apps deliver consistent results across different environments.

The original TNA implementation is provided through *tna* [29], which offers transition-network estimation, centrality analysis, bootstrapping, permutation testing, case-dropping procedures, and the visual conventions used in published TNA research. Complex-network analysis and rendering are handled through *cograph* [30], which provides community detection, multilayer and heterogeneous network visualisation, centrality analysis, and a broad suite of plotting methods including transition, comparison, bootstrap, heatmap, trajectory, and alluvial plots. Cograph functions as the dedicated rendering layer and operates independently from the estimation engine through object-based dispatch.

At the core of the framework is *Nestimate* [18], the main estimation and inference engine. Nestimate implements the major computational components of the framework, including transition-network estimation, bootstrap and permutation inference, split-half reliability, case-dropping procedures, centrality stability, network comparison testing, Markov-order analysis, path-dependence diagnostics, mixed Markov models, multilevel VAR, mixed graphical models, higher-order networks, hypergraphs, and simplicial analyses. Its architecture is based on a registry and dispatch system in which estimators are accessed through a unified interface and custom methods can be integrated alongside built-in estimators. Documentation is available online.

Specialised extensions support additional methodological families. The *htna* package provides Heterogeneous Transition Network Analysis (HTNA) for modelling sequences involving multiple interacting actor groups such as humans and AI systems. The *cooccure* package implements Co-occurrence Network Analysis with multiple similarity normalisations and flexible data input structures for bipartite, binary, sequence, and delimited formats [31].

A defining feature of the software architecture is the strict separation between computation, inference, and rendering. Estimation packages generate shared netobject structures and specialised subclasses, while downstream analytical and visualisation functions dispatch behaviour based on object class rather than direct package dependencies. This modular architecture allows methods, interfaces, and rendering systems to evolve independently while preserving a coherent analytical framework across the entire *Dynalytics* ecosystem.

The aforementioned packages have been originally developed in the R programming language. However, alternatives in other common languages exist as well as graphical interfaces for non-technical users. For instance, *tna* is also offered for the Python language (*tnapy*) [32]. TNA is also available as a web app based on R Shiny [33] as well as for desktop as part of the Jamovi app [34]. A newer, more comprehensive web application that combines the complete *Dynalytics* framework is also available online. Dynalytics has also been implemented from scratch in JavaScript (*dynajs*) and can be used to implement real-time dashboards.

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